

SANTHOSH KUMAR VINAYAGAM

ADAS ECU Software Architect | AUTOSAR Classic | ISO 26262 (ASIL-B/D)

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PROFESSIONAL SUMMARY

ADAS ECU Software Architect with 12+ years of experience designing and delivering safety-critical embedded systems for global automotive OEMs. Extensive onsite experience in the United States leading Driver Monitoring System (DMS) ECU architecture, AUTOSAR integration, and vehicle-level validation.

Expert in AUTOSAR Classic stack design, BSW integration, safety lifecycle execution (ISO 26262), and OEM-facing technical leadership. Proven ability to architect scalable ECU platforms, optimize system performance, and lead cross-functional global teams across the full V-cycle.

CORE ARCHITECTURE COMPETENCIES

- AUTOSAR Classic Architecture (MCAL, BSW, RTE, OS, COM Stack)
- ECU Software Partitioning & Integration Strategy
- ISO 26262 Functional Safety Lifecycle (ASIL-B / ASIL-D) – Safety Concept, FMEA, FTA
- ADAS ECU Design – Driver Monitoring Systems
- Communication Protocols (CAN, CAN-FD)
- Bootloader & Secure OTA Update Architecture
- Cybersecurity Architecture (Secure Boot, Encryption, Key Management – ISO 21434)
- Requirements & Architecture Modeling (Enterprise Architect, MATLAB/Simulink)
- OS Configuration & Task Scheduling
- Diagnostic Architecture (UDS / ISO 14229)
- Memory & Performance Optimization
- SIL / HIL / MIL Validation Strategy
- ASPICE SWE.1–SWE.6 Compliance

PROFESSIONAL EXPERIENCE

HCL Technologies Ltd | 2014 – Present

ADAS ECU Software Architect – Driver Monitoring System (ASIL-B, AUTOSAR Classic)

Client: North American Automotive OEM | Offshore (2023 till date) | Onsite – Detroit, MI, USA (2017-2022)

Architecture & Technical Ownership

- Led AUTOSAR Classic-based ECU architecture for camera-driven Driver Monitoring System supporting drowsiness and distraction detection.
- Owned BSW stack integration including MCAL, Port, SPI, I2C, OS, COM, and Diagnostic stack configuration.
- Defined ECU software partitioning, RTE configuration, and OS task mapping aligned with ASIL-B safety goals.
- Designed integration strategy between application SW-Cs, middleware, and hardware abstraction layers.

Safety & Compliance

- Executed ISO 26262 software lifecycle activities including safety requirements allocation, traceability, and audit readiness.
- Reduced static analysis violations by 35% through structured remediation and compliance strategy.
- Supported safety reviews, FMEA inputs, and release readiness assessments.

Diagnostics & Vehicle Integration

- Architected UDS diagnostic services supporting flashing, calibration, and production workflows.
- Led vehicle-level ECU integration and calibration support during validation and pre-SOP milestones.
- Acted as primary technical interface between OEM system teams, Tier-1 suppliers, and offshore development teams.

Senior AUTOSAR Configuration Engineer- AUTOSAR ASW and BSW configuration

Client: German OEM | 2016-2017

- AUTOSAR configuration using tools like Vector DaVinci Configurator/Developer and EB Tresos
- End-to-end BSW configuration (NvM, DCM, DEM)
- Migration of legacy (non-AUTOSAR) software to AUTOSAR-based architecture including SWC redesign and integration
- SWC design and configuration – Ports, Interfaces (Sender-Receiver / Client-Server), Data Elements
- RTE generation and Runnable configuration (timing events, data events, operation invoked events)
- ECU extract, system configuration, and ARXML handling (System/ECU/BSWMD)
- Communication stack configuration (CAN / CAN-FD – signal mapping, PDU routing)
- Integration and validation of ASW-BSW interaction across layers
- Memory stack configuration (NvM, Fee) and optimization
- Diagnostic stack configuration (UDS – DCM, DEM, DID, DTC handling)
- Implemented secure diagnostic access (UDS Security Access – Seed/Key mechanism)
- Supported secure flashing / OTA update mechanisms ensuring authenticity and rollback protection

Powertrain ECU Validation Engineer – Engine Control Module

2015-2016

- Designed and executed Hardware-in-the-Loop validation campaigns.
- Developed Design Verification Plans (DVP) and traceability matrices aligned with V-model.
- Automated regression testing workflows improving validation efficiency.

Embedded UI & Diagnostics Developer – Steering Column Lock System 2014-2015

- Developed MATLAB/Simulink based diagnostic and validation GUI tools.
- Integrated hardware interfaces and safety-aligned test procedures under ISO 26262 framework.

Project Assistant Engineer – Wind Turbine Modeling (MATLAB/Simulink) Centre for Wind Energy Technology (C-WET) | 2012 – 2014

- Developed MATLAB/Simulink models for wind turbine systems covering aerodynamic, mechanical, and control domains for performance analysis and validation.
- Designed and validated control logic for turbine behavior under varying wind conditions, improving model stability and accuracy.

TECHNICAL EXPERTISE

Programming: Embedded C, C++, CAPL, Python

AUTOSAR Tools: DaVinci Configurator Pro, DaVinci Developer, EB Tresos

MCUs: NXP MPC55xx, S32K312

Standards: AUTOSAR, ISO 26262, ASPICE, MISRA

CI/CD & Configuration Management: Jenkins, SVN, Serena

LEADERSHIP & IMPACT

- 5 years direct OEM onsite engagement in US automotive program.
- Led cross-functional collaboration across architecture, validation, and system teams.
- Supported ECU releases from development through vehicle validation phases.
- Strengthened software quality through systematic static analysis compliance improvements.
- Contributed to global onsite–offshore delivery execution model.

EDUCATION

Bachelor of Engineering – Electrical & Electronics Engineering (2007-2011)